

SPECIAL EVENTS

PHASES OF THE MOON

	FULL MOON	NEW MOON
2020	June 5	June 21
2021	June 24	June 10
2022	June 14	June 29
2023	June 4	June 18
2024	June 22	June 6
2025	June 11	June 25
2026	June 30	June 15
2027	June 19	June 4

PLANETS

- 2020: June 4** Mercury is at greatest evening elongation, magnitude 0.7.
- 2022: June 16** Mercury is at greatest morning elongation, magnitude 0.7.
- 2023: June 4** Venus is at greatest evening elongation, magnitude -4.3
- 2023: June 4** Venus is at greatest morning elongation, magnitude -4.3.
- 2026: June 15** Mercury is at greatest evening elongation, magnitude 0.7.
- 2028: June 26** Mercury is at greatest morning elongation, magnitude 0.7.

ECLIPSES AND TRANSITS

- 2020: June 21** An annular eclipse of the Sun is visible from central Africa, south Asia, China, and the Pacific.
- 2021: June 10** An annular eclipse of the Sun is visible from north Canada, Greenland, and Russia.

JUNE

Northern nights are at their shortest, and southern ones at their longest, around the solstice on June 21, the date on which the Sun reaches its farthest point north of the celestial equator. In the northern sky, Arcturus and the other stars of Boötes stand high, and the giant Summer Triangle of Vega (in Lyra), Deneb (in Cygnus), and Altair (in Aquila) lies in the eastern half of the sky. Southern observers enjoy a rich band of constellations in the Milky Way during their long winter nights.

NORTHERN LATITUDES

THE STARS

The bowl of the Little Dipper, in Ursa Minor, stands high above the northern horizon with the sinuous body of Draco, the Dragon, winding around it. The horseshoe shape of Corona Borealis, the Northern Crown, lies on the north-south meridian with the head of Serpens, the Serpent, below it, while Arcturus, in Boötes, is high in the western half of the sky. In this area of sky, Arcturus is the base of a large Y-shaped pattern of bright stars completed by Epsilon (ϵ) and Gamma (γ) Boötis plus Alpha (α) Corona Borealis (also known as Alphecca). Leo is setting in the west,

and Spica, in Virgo, is low in the southwest. In the eastern sky, the bright stars Vega (see p.253), Deneb, and Altair (see p.252) mark the corners of the Summer Triangle, best seen in late summer and fall. Ruddy Antares and the stars of Scorpius twinkle low on the southern horizon—June and July are the best months of the year for far-northern observers to see Scorpius in the evening sky.

DEEP-SKY OBJECTS

The brightest globular cluster in northern skies, M13, is high up on summer evenings. It can be found along one side of the Keystone of Hercules, a quadrangle of stars that forms the torso of the constellation Hercules. M13 appears as a fuzzy

NOCTILUCENT CLOUDS

These high-altitude clouds can be seen on summer nights, illuminated by the Sun's rays that come over the horizon around midnight.

6th-magnitude star through binoculars, and it can be glimpsed by the naked eye under good conditions. It can be compared to M5, another 6th-magnitude globular cluster visible through binoculars. M5 lies in the head of Serpens and is usually regarded as the second-best northern globular cluster. Near the handle of the Big Dipper, the spiral galaxies M51 and M101 remain well positioned for observation.

